THE LOWER SHRIEK COMPUTED BY THE STANDARD RESOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof environment carries the author's optional label 'First proof'; the optional argument is part of the counted proof span and is preserved verbatim.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the lower shriek by polynomial resolutions with a trivial Kan fibration. Lemma lemma-identify-pi-shriek (445-456, produced as L44140), used at proof lines 506-510. Reduces the claim to its two hypotheses.
- The standard resolution map is a homotopy equivalence. Simplicial, Lemma simplicial-lemma-standard-simplicial-homotopy, used at proof lines 510-513. Gives the first step of the chain.
- A homotopy equivalence induces a quasi-isomorphism of associated complexes. Simplicial, Lemma simplicial-lemma-homotopy-equivalence, used at proof lines 513-516. Gives the second step.
- A quasi-isomorphism of associated complexes makes the map a trivial Kan fibration. Simplicial, Lemma simplicial-lemma-qis-simplicial-abelian-groups, used at proof lines 516-519. Concludes the second hypothesis.

Source reference. Original equation 08PR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11034–11034. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11035–11035. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11010–11010. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08P1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11009–11009. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08ND: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10986–10986. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $\epsilon : P_\bullet \to B$ be the standard resolution of B over A. Let π be as in (08PR). Then*

$$L\pi_!(\mathcal{F}) = \mathcal{F}(P_\bullet, \epsilon)$$

in $D(Ab)$, resp. $D(B)$ functorially in $\mathcal{F}$ in $Ab(\mathcal{C})$, resp. $Mod(\underline{B})$.

First proof. We will apply Lemma 08PS. Since the terms P_n are polynomial algebras we see the first assumption of that lemma is satisfied. The second assumption is proved as follows. By Simplicial, Lemma 08ND the map ϵ is a homotopy equivalence of underlying simplicial sets. By Simplicial, Lemma 08P2 this implies ϵ induces a quasi-isomorphism of associated complexes of abelian groups. By Simplicial, Lemma 08P1 this implies that ϵ is a trivial Kan fibration of underlying simplicial sets. □